

BLYTHEVILLE, AR, USA

WMO: 723409

Lat: 35.940N

Lon: 89.831W

Elev: 78

StdP: 100.40

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 53869

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-6.3	-14.6	1.1	-6.8	-12.3	1.3	-4.6	12.7	6.3	11.8	6.5	4.5	20	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	35.2	25.1	34.1	24.8	33.0	24.5	27.0	32.4	26.4	31.9	25.7	30.9	3.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.6	21.0	30.3	24.8	20.1	29.5	24.2	19.3	28.9	85.1	32.4	82.3	31.8	79.6	31.1	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.3	8.3	-11.9	37.0	3.4	1.5	-14.3	38.1	-16.3	39.0	-18.2	39.9	-20.7	41.0		
			-12.7	28.3	3.1	0.8	-14.9	28.9	-16.7	29.3	-18.4	29.8	-20.7	30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	3.9	5.4	10.8	16.5	21.7	26.4	27.1	26.3	22.9	16.8	10.5	5.8
	DBStd	9.57	6.18	5.96	5.82	4.87	4.15	2.62	2.38	2.65	3.64	4.95	5.25	5.16
	HDD10.0	643	207	153	61	8	0	0	0	0	0	5	57	152
	HDD18.3	1890	449	364	241	91	20	0	0	0	7	90	240	389
	CDD10.0	2913	17	23	86	202	363	492	531	506	386	216	71	21
	CDD18.3	1117	0	1	7	34	124	242	272	247	143	42	4	0
	CDH23.3	11221	0	1	41	256	1044	2625	2985	2541	1320	385	23	1
	CDH26.7	4738	0	0	4	48	356	1194	1343	1143	533	115	2	0
Wind	WSAvg	3.7	4.6	4.6	4.6	4.6	3.9	3.3	2.6	2.4	2.6	3.3	3.9	4.2
Precipitation	PrecAvg	1297	105	109	124	130	130	103	109	73	85	102	101	130
	PrecMax	1640	221	222	275	280	276	199	268	135	171	259	233	321
	PrecMin	968	43	41	30	34	24	37	21	15	10	0	25	24
	PrecStd	184	42	48	59	60	61	40	64	42	45	61	54	69
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	22.9	27.1	30.0	33.2	36.3	36.1	37.2	35.0	31.9	26.2	21.2
		MCWB	17.6	18.1	18.6	20.1	22.5	24.2	25.6	25.4	23.6	22.3	18.9	16.8
	2%	DB	18.7	20.2	24.1	28.0	31.6	34.7	34.7	35.1	33.2	29.7	23.2	18.4
		MCWB	16.5	16.4	17.6	19.2	21.8	24.0	25.7	25.8	23.9	21.3	17.1	15.6
	5%	DB	16.4	17.6	22.1	26.4	30.2	33.6	33.7	33.6	31.8	27.5	21.2	16.3
		MCWB	13.8	13.9	16.4	18.3	21.3	23.6	25.4	25.5	23.5	19.5	16.1	13.6
	10%	DB	13.6	15.1	20.0	24.4	28.8	32.4	32.6	32.3	30.0	25.2	18.9	14.0
		MCWB	10.9	12.0	14.9	17.8	20.9	23.3	25.1	25.2	22.5	18.6	14.9	11.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.2	19.1	19.8	22.1	24.5	26.6	27.9	28.1	25.9	24.0	20.8	18.2
		MCDB	19.9	21.8	24.3	27.2	29.7	33.0	33.2	34.0	31.9	28.8	23.6	20.6
	2%	WB	16.7	17.2	18.5	20.8	23.5	25.7	27.0	27.2	25.1	22.7	18.8	16.2
		MCDB	18.5	20.0	22.9	25.8	28.7	31.9	32.3	32.7	30.9	27.7	22.2	18.1
	5%	WB	14.6	14.9	17.2	19.9	22.8	25.1	26.5	26.5	24.3	21.5	17.2	14.0
		MCDB	16.2	17.2	21.3	24.5	28.0	31.0	32.0	31.8	29.6	25.7	19.7	15.9
	10%	WB	11.4	12.2	15.8	18.8	22.1	24.5	25.8	25.7	23.6	20.1	15.4	12.0
		MCDB	13.3	14.6	19.4	23.2	27.1	30.0	30.9	30.7	28.3	23.9	18.5	13.7
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.0	10.3	10.8	10.0	10.1	9.5	10.5	11.8	11.9	10.7	8.5
		MCDBR	10.7	11.9	12.0	11.9	11.3	11.3	10.4	11.8	12.9	13.7	12.9	10.5
		MCWBR	9.1	8.9	7.4	5.7	4.4	3.7	3.8	4.6	5.1	6.3	8.0	8.3
	5% WB	MCDBR	9.9	10.6	10.8	10.2	9.3	9.9	9.5	10.2	10.8	11.2	10.3	9.4
		MCWBR	9.2	8.8	7.2	6.0	4.4	4.0	4.2	4.8	4.8	6.0	8.2	8.4
Clear-Sky Solar Irradiance	taub	0.316	0.328	0.366	0.406	0.438	0.454	0.485	0.483	0.418	0.358	0.326	0.315	
	taud	2.486	2.442	2.369	2.256	2.230	2.187	2.139	2.126	2.302	2.452	2.492	2.497	
	Ebn at Noon	883	911	901	878	851	834	805	798	837	866	863	859	
	Edh at Noon	83	97	115	135	141	147	153	152	121	95	82	78	
All-Sky Solar Radiation	RadAvg	2.36	2.90	4.01	5.18	5.81	6.51	6.29	5.75	4.86	3.74	2.61	2.01	
	RadStd	0.20	0.38	0.34	0.35	0.42	0.40	0.31	0.33	0.32	0.43	0.27	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page